PARSHWANATH CHARITABLE TRUST'S
A.P. SHAH INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Data Science

Lineless

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Project Guide

Ms. Sarala Mary

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1. Introduction

Problem Identified :

- Students waste time standing in long queues for services like Scholarship, Train Pass, and Hostel Issue.
- Manual token systems cause confusion, delays, and lack of transparency.
- No proper way to track progress or notify users when their turn comes.

Solution Proposed :

- A Digital Token Management System built using Flutter + Firebase.
- Tokens are generated digitally and arranged category-wise using database filter queries.
- Each request gets a unique token number stored in Firestore.
- Users and admins can track status, mark completion, and maintain history.
- Admins can manage breaks in queues with backend logic, ensuring smooth operation.

2. Objectives

- To provide category-wise queues by using database filter queries, so tokens are arranged separately for categories like Scholarship, Train Pass, or Hostel Issue.
- To generate a unique token for each user request by using Flutter forms for input and auto-increment fields in the database to store and display the token number.
- To allow users to cancel their token if they no longer need the service, while only the admin has the authority to mark a token as complete in the database.
- To manage admin-defined breaks in queues by using a breaks table in the database and time comparison logic in the backend, so users are notified when the queue is paused and when it resumes.

3. Scope

1. Can be applied in educational institutions like colleges and universities to manage student service queues (Scholarship, Train Pass, Hostel Issue).
2. Can be useful to students by reducing waiting time, providing transparency, and notifying them when their turn comes.
3. Can be useful to administrators by simplifying queue management, tracking completed requests, and handling breaks efficiently.
4. Can be extended to other domains such as hospitals, government offices, or banks where multiple service queues need to be managed digitally.

4. Feature /Functionality

Digital Token Generation

- Users can generate tokens digitally for different queues such as Scholarship, Train Pass, or Hostel Issue.
- Tokens are stored in the database with unique IDs and arranged category-wise for clarity.

Estimated Time & Token Control

- The system shows the estimated waiting time for a user's turn in the queue.
- Users can cancel their token anytime if they no longer need the service, while only admins can mark tokens as complete.

Student Details Page

- A dedicated student page stores all important documents and personal details securely.
- This ensures easy access for both students and admins when verifying requests

5. Outcome of Project

User can generate a digital token through Flutter forms, which is stored in the database with a unique number.

User can view category-wise queues (Scholarship, Train Pass, Hostel Issue) arranged separately using database filter queries.

Admin can manage breaks in queues using backend time comparison logic, and users are notified when queues pause or resume.

System maintains a history of completed tokens for tracking and transparency.

6. Technology Stack

- Flutter
- Firebase
- Flask

7. Frontend & Backend

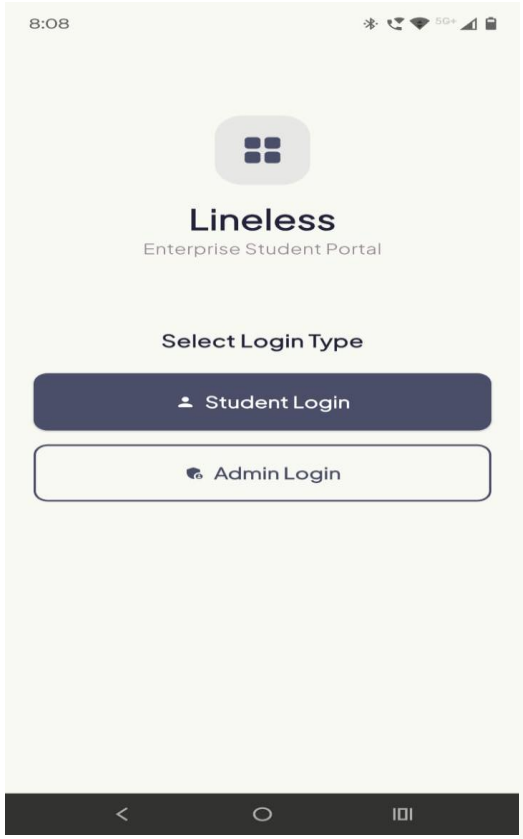
```
Lineless > linelss > lib > main.dart > baseUrl
class LoginSelectionScreen extends StatelessWidget {
  const LoginSelectionScreen({super.key});

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      body: SafeArea(
        child: Padding(
          padding: const EdgeInsets.all(24.0),
          child: Column(
            crossAxisAlignment: CrossAxisAlignment.stretch,
            children: [
              const SizedBox(height: 60),
              Center(
                child: Container(
                  width: 80,
                  height: 80,
                  decoration: BoxDecoration(
                    color: LinelessTheme.primary.withOpacity(0.1),
                    borderRadius: BorderRadius.circular(24),
```

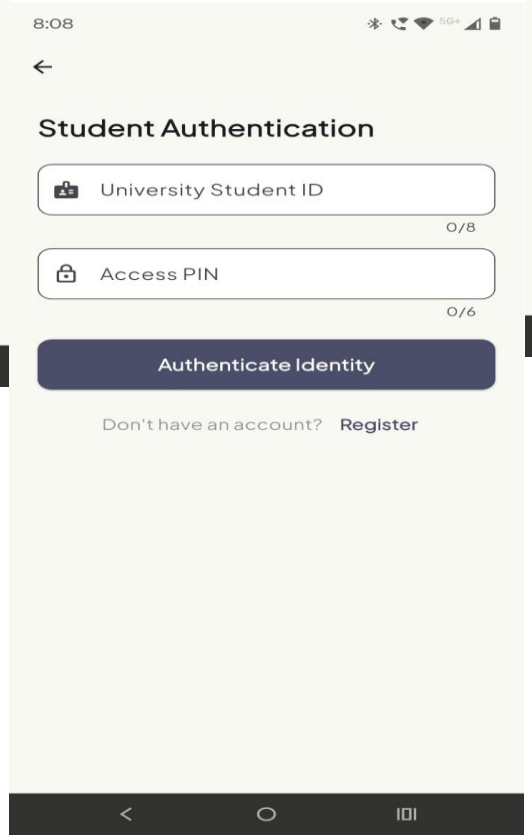
```
PROBLEMS 10   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

 * Debugger is active!
 * Debugger PIN: 177-388-394

C:\Users\Nikhil\OneDrive\Desktop\Mini Project\Lineless\linelss\backend>C:\Users\Nikhil\anaconda3\Library\bi
n\conda.BAT activate base
```

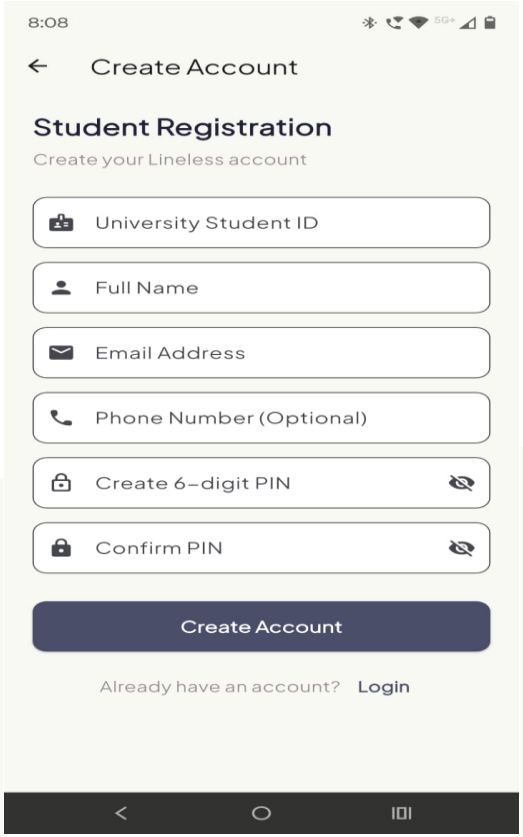
```
Lineless > linelss > backend > app.py > ...
    init_notification_service(app.config.get('FIREBASE_CREDENTIALS_PATH'))

# Token counter (for generating unique token IDs)
token_counter = 88290

def update_queue_positions(service_id):
    """Update queue positions for all active tokens of a service"""
    try:
        active_tokens = Token.query.filter_by(
            service_id=service_id,
            status='ACTIVE'
        ).order_by(Token.created_at).all()

        notification_service = get_notification_service()

        for idx, token in enumerate(active_tokens):
            old_position = token.queue_position
            new_position = idx + 1
            token.queue_position = new_position
```

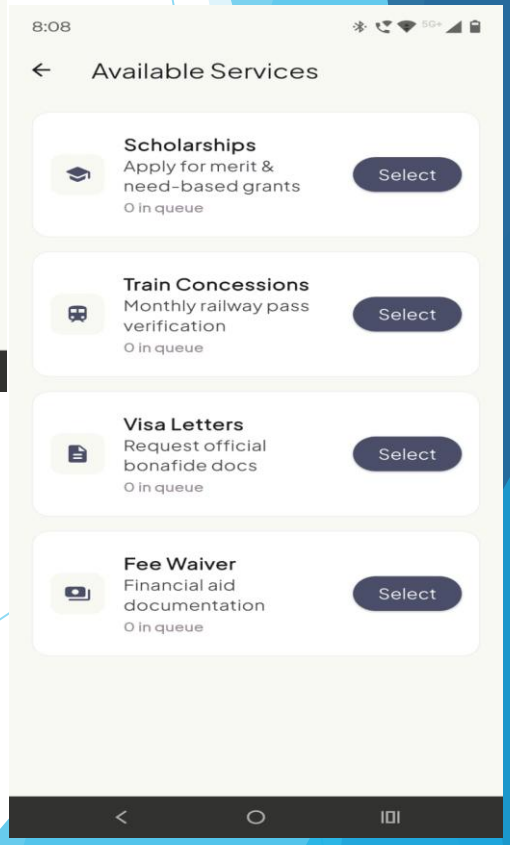
```
PROBLEMS 10   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

 * Debugger is active!
 * Debugger PIN: 177-388-394

C:\Users\Nikhil\OneDrive\Desktop\Mini Project\Lineless\linelss\backend>C:\Users\Nikhil\anaconda3\Library\bi
n\conda.BAT activate base

(base) C:\Users\Nikhil\OneDrive\Desktop\Mini Project\Lineless\linelss\backend>
```

8. Output:-

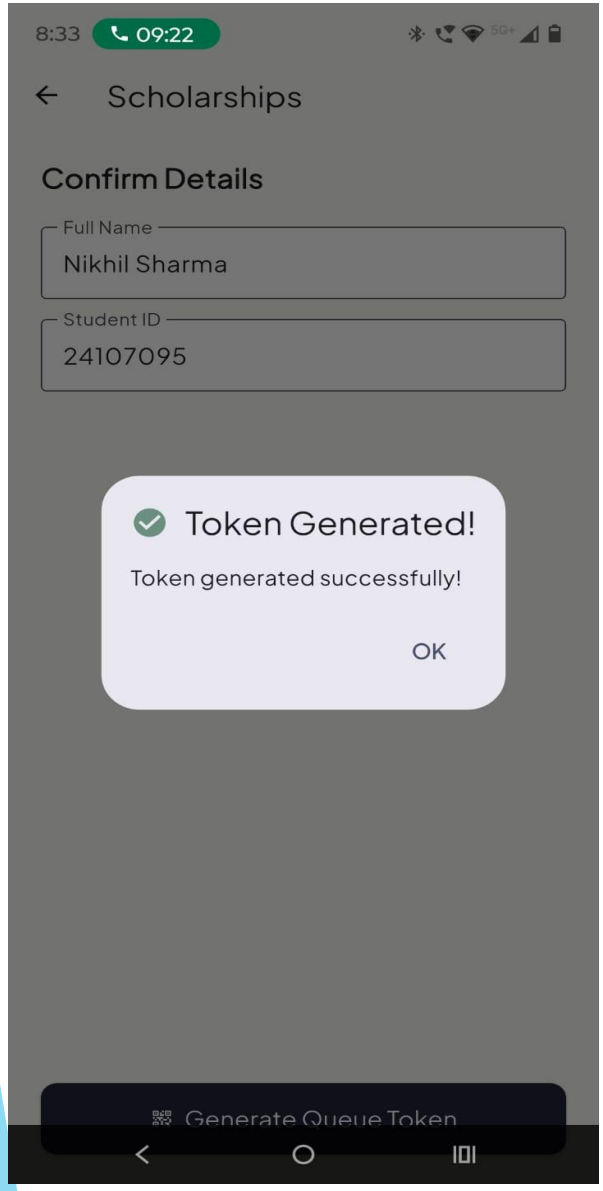

8:08
Lineless
Enterprise Student Portal
Select Login Type
Student Login
Admin Login

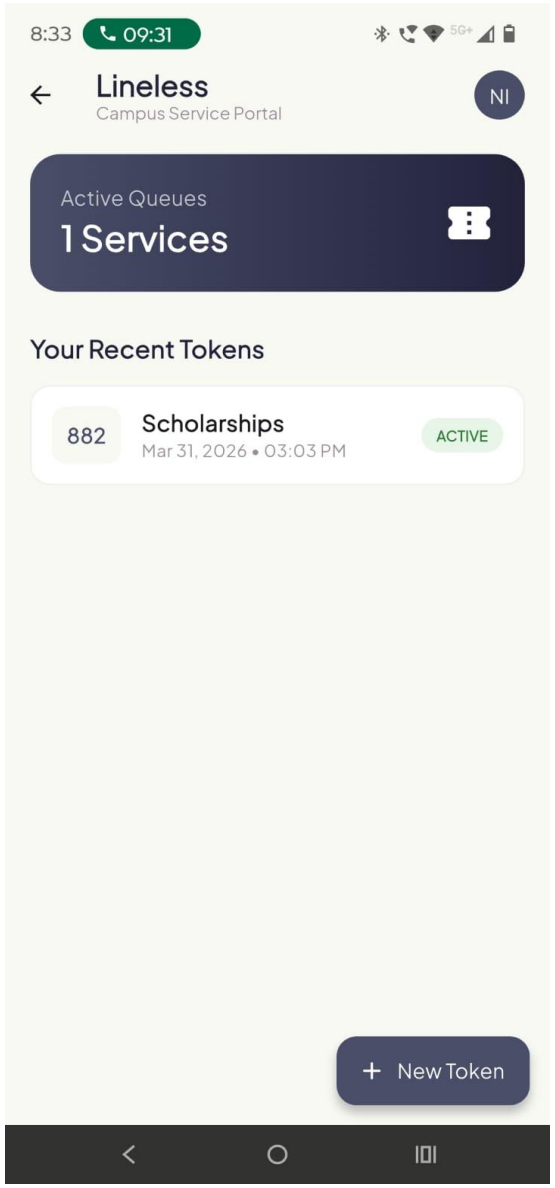

8:08
Student Authentication
University Student ID
0/8
Access PIN
0/6
Authenticate Identity
Don't have an account? Register

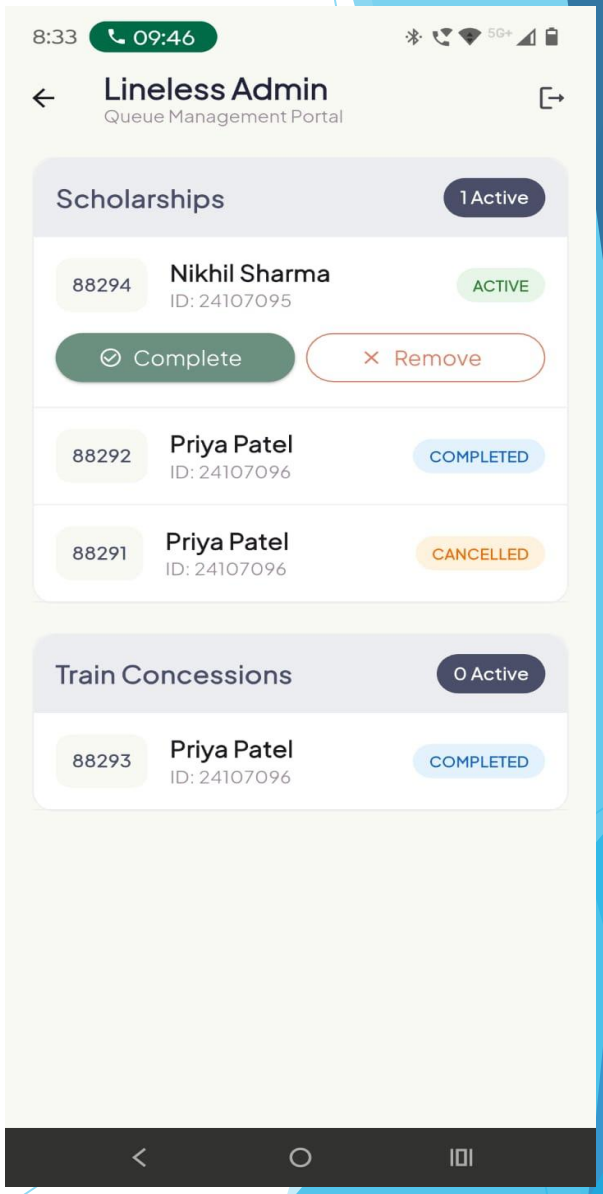

8:08
Create Account
Student Registration
Create your Lineless account
University Student ID
Full Name
Email Address
Phone Number (Optional)
Create 6-digit PIN
Confirm PIN
Create Account
Already have an account? Login


8:08
Available Services
Scholarships
Apply for merit & need-based grants
0 in queue
Select
Train Concessions
Monthly railway pass verification
0 in queue
Select
Visa Letters
Request official bonafide docs
0 in queue
Select
Fee Waiver
Financial aid documentation
0 in queue
Select

8. Output:-


8:33
09:22
Scholarships
Confirm Details
Full Name
Nikhil Sharma
Student ID
24107095
Token Generated!
Token generated successfully!
OK
Generate Queue Token


8:33
09:31
Lineless
Campus Service Portal
NI
Active Queues
1 Services
Your Recent Tokens
882
Scholarships
Mar 31, 2026 • 03:03 PM
ACTIVE
+ New Token


8:33
09:46
Lineless Admin
Queue Management Portal
Scholarships
1 Active
88294
Nikhil Sharma
ID: 24107095
ACTIVE
Complete
Remove
88292
Priya Patel
ID: 24107096
COMPLETED
88291
Priya Patel
ID: 24107096
CANCELLED
Train Concessions
0 Active
88293
Priya Patel
ID: 24107096
COMPLETED

Thank You...!!